

Drone Ballet

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1 Convection Values

The goal of this section is to outline the process used to determine the convection coefficient as a function of atmospheric conditions and drone rotation rate. All methods were laid out by Theodore L. Bergman in **Fundamental of Heat and Mass Transfer - Eighth Edition**. First, the ideal gas law was used to determine the density of air as a function of temperature and pressure:

- $M = 0.02987 \text{ kg/mol}$
- $P = 101325 \text{ Pa}$
- $T = 273.15 \text{ K}$
- $R = 8.314 \text{ J/(mol} \cdot \text{K)}$

Rearranging the ideal gas law gives the following expression for air density:

$$\rho = \frac{PM}{RT}$$

Using the values above gives

$$\rho = \frac{(101325)(0.02987)}{(8.314)(273.15)} = 1.3327 \text{ kg/m}^3.$$

The following fluid properties of air were then used:

- $\mu = 1.86 \times 10^{-5} \text{ Pa} \cdot \text{s}$
- $\text{Pr} = 0.711$
- $k = 0.02436 \text{ W/(m} \cdot \text{K)}$

To simplify the heat transfer and fluid dynamics calculations and improve computational efficiency, the drone was approximated as a cylinder. The major axis of the drone was therefore used as the characteristic diameter of the cylinder:

$$D = 0.3393 \text{ m.}$$

The drone rotation rate was varied from 0 to 360 degrees/s. For each rotation rate, the corresponding angular velocity was first converted to radians per second:

$$\omega = \dot{\theta} \frac{\pi}{180},$$

where $\dot{\theta}$ is the drone rotation rate in degrees per second.

The tangential velocity at the outer surface of the equivalent cylindrical body was then calculated as

$$v = \omega \frac{D}{2}.$$

Using this tangential velocity, the Reynolds number was calculated according to

$$\text{Re}_D = \frac{\rho v D}{\mu}.$$

The Nusselt number for flow around a cylinder was then calculated using the following discrete Reynolds number ranges:

- $0.4 < \text{Re}_D < 4$: $C = 0.989$, $m = 0.385$
- $4 < \text{Re}_D < 40$: $C = 0.911$, $m = 0.385$
- $40 < \text{Re}_D < 4000$: $C = 0.683$, $m = 0.466$
- $4000 < \text{Re}_D < 40000$: $C = 0.027$, $m = 0.805$

For each Reynolds number range, the Nusselt number was calculated using

$$\text{Nu}_D = C \text{Re}_D^m \text{Pr}^{1/3}.$$

Finally, the convection coefficient was determined from the Nusselt number using

$$h = \frac{\text{Nu}_D k}{D},$$

where h is the convection coefficient in $\text{W}/(\text{m}^2 \cdot \text{K})$.

This process was repeated for each drone rotation rate to determine the corresponding convection coefficient as a function of rotation speed.